

Football Data Architecture Documentation

Data Lake House Implementation on AWS for Advanced Analytics

Project developed for

Federación Mexicana de Fútbol (FMF)

Implemented by

Team 6 - Data Engineering, UNAM 2026.

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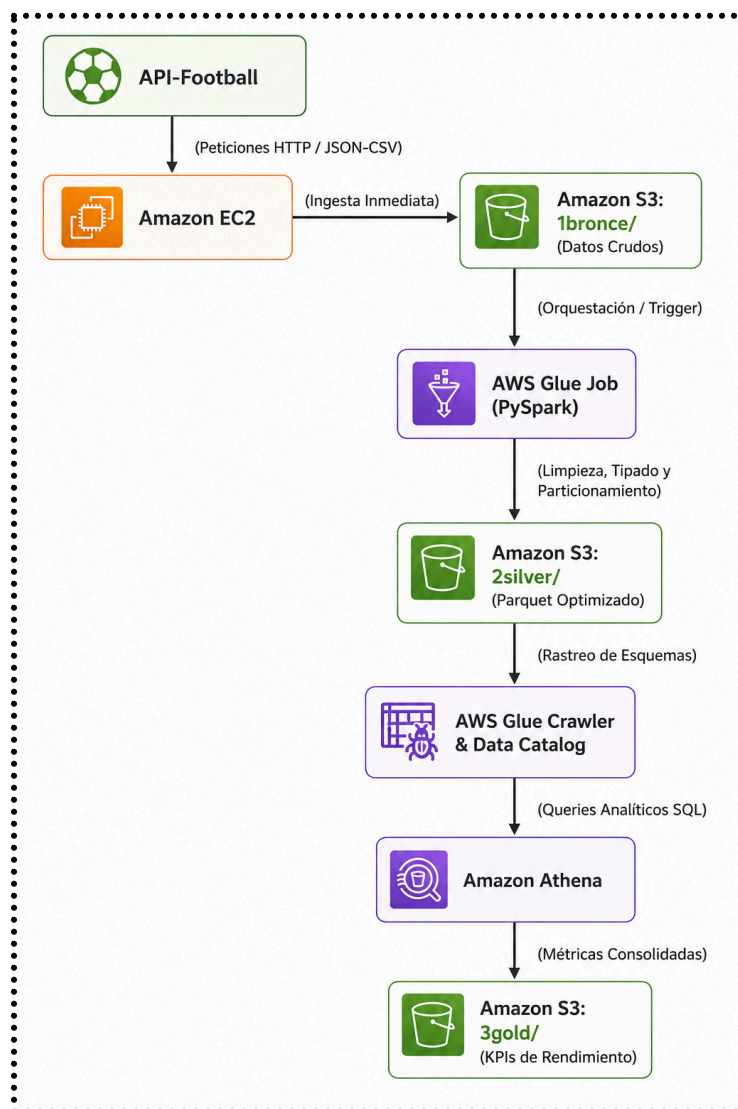
1. Executive Summary

The objective of the system is to perform ingestion, cleansing, transformation, and analysis of historical match data for rival national teams, considering the proximity of the 2026 World Cup.

The solution implements a Data Lakehouse architecture under the Medallion pattern (Bronze, Silver, Gold) using fully managed services on Amazon Web Services (AWS). It enables the transformation of raw data from external sources into competitive advantages and highly reliable sporting performance metrics (KPIs) for tactical decision-making.

2. Architecture Diagram (Conceptual)

The architecture follows a decoupled lineage where storage and compute scale independently:



3. Technology Stack

To ensure resilience, distributed processing speed, and a Serverless environment that minimizes operational costs, the following components from the AWS ecosystem were used:

- **Data Sources:**

API-Football, (external provider of historical international football match statistics, consumed via HTTP requests in JSON/CSV format)

- **Ingestion Service, Amazon EC2 (Elastic Compute Cloud):**

A virtual instance responsible for running scheduled Python extraction scripts to consume the external API.

- **Centralized Storage (Data Lake):**

Amazon S3 (Simple Storage Service), the storage core, hierarchically segmented into buckets/folders for the data maturity layers (Bronze, Silver, Gold)

- **Big Data Processing (ETL), AWS Glue Jobs:**

A Serverless Spark engine optimized to run large-scale distributed PySpark transformations without the need to manage physical clusters.

- **Catalog and Governance, AWS Glue Data Catalog & Crawlers:**

Responsible for automatically inferring the schema of optimized structures and exposing them as logical tables.

- **Consumption and Analytics Layer, Amazon Athena:**

A Serverless interactive query engine that allows executing standard SQL instructions directly on files stored in Amazon S3.

4. Layer Description (Medallion Architecture)

Layer	Process Description	Technology
Bronze (Raw)	Hosts data in its original format. No transformation is applied, preserving the pure historical record for audits or reprocessing	Amazon S3 / CSV files per national team
Silver (Trusted)	Contains clean and enriched data. Strict type transformations, null elimination, and text string standardization are applied. Data is converted to a format optimized for analytics.	Amazon S3 / Columnar storage format (Parquet) natively partitioned by year.
Gold (Refined)	Final consumption layer. Stores aggregated views and sporting business metrics calculated from the Silver layer. Optimized to answer specific questions.	Amazon S3 / Structured table queries via Amazon Athena

5. Data Flow (Pipeline)

1. Extraction and Ingestion:

An Amazon EC2 instance runs scheduled Python scripts that make HTTP requests to the API-Football endpoint, downloading the match history of each national team in CSV/JSON format. The extracted files are deposited directly into the Bronze Layer of

the centralized bucket

s3://unam-2026-ingenieriadatos-equipo6/1bronce/

2. Distributed Processing (ETL):

An AWS Glue job loads raw data into a Spark DataFrame. The script runs the following data quality tasks in a distributed manner:

- Removal of null records in critical control fields
- Text standardization (whitespace cleansing via trim and uppercase conversion with upper)
- Formatting the time column to Timestamp type
- Casting home and away goals to Integer type

3. Optimization and Writing:

Clean data is exported to the Silver layer path using Parquet format. To optimize analytical searches and drastically reduce scan times, physical partitioning by year is applied, storing structures in isolation per national team

4. Metadata Discovery:

An AWS Glue Crawler periodically scans the Silver layer. Upon detecting new partitioned folders, it updates the AWS Glue Data Catalog, synchronizing the tables for immediate querying.

5. Insights Generation (Gold Layer):

Through Amazon Athena, the interested client runs analytical SQL queries to calculate advanced metrics (win percentage, average goals scored and conceded per match, historical performance in specific contexts), ready for strategic consumption.

6. Security Considerations

• Least Privilege Policies:

Access to resources is restricted through AWS IAM (Identity and Access Management) roles, strictly limiting the write permissions of the Glue Job exclusively to its corresponding buckets

- **Cost Efficiency:**

The use of Parquet and partitioned queries in Athena ensures that only the data blocks required by the active query are scanned, reducing AWS computing costs by more than 80% compared to scanning plain text files